Bond Sphere: Diatomic Molecules as S^3 and their Natural Sturmian Basis*

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A diatomic molecule is geometrically a single three-sphere S^3 with two weighted poles—the *bond sphere*—in which the internuclear distance R encodes as a polar angle γ via $\cos\gamma = (p_0^2 - p_R^2)/(p_0^2 + p_R^2)$ and all cross-atom matrix elements are $SO(4)$ Wigner D-matrix elements evaluated at the bond rotation $e^{i\gamma A_y}$. *Positive results:* The bond sphere geometry is consistent, with the $\gamma(R)$ encoding, heteronuclear egg metric, and Runge–Lenz alignment following from $SO(4)$ representation theory. (A graph-partition "kinetic membrane" is also described; it is built on the graph Laplacian, which is not $SO(4)$-covariant, and its Fiedler-shift and Bader-analog claims are unverified.) The D-matrix enforces selection rules for σ-bonds: $D^2_{10,10}(\gamma) = 1$ and $D^2_{00,10}(\gamma) = 0$ for all γ. *Negative theorem:* In the molecular Sturmian basis with shared momentum scale p_0, the one-electron Hamiltonian and overlap obey the two-term identity $H_{ij} = -\frac{1}{2}p_0^2 S_{ij} + (1-\beta_j)V_{ij}$, and their entire R-dependence is a single orthogonal $SO(4)$ congruence $D^{(n)}(\gamma(R))$ that cancels in the generalized eigenvalue problem: the eigenvalues are independent of R. In this single-n, shared-p_0 form, binding would require the R-dependent prolate spheroidal spectrum $\beta_k(R)$; but the genuine cross-n Coulomb–Sturmian overlap couples n, escapes the theorem, and supplies that R-dependence algebraically via Shibuya–Wulfman (Remark 1), so what is ruled out is this approximation, not Sturmian molecular binding. The continuous-geometry route—working in the label set and channel structure of prolate spheroidal coordinates, where the problem separates—is developed in the companion Paper 11 [19].

I. INTRODUCTION

In the GeoVac framework [1–3], the quantum numbers (n, l, m) label the harmonics of the unit three-sphere S^3. The principal quantum number n indexes shells, each containing n^2 labels; the orbital angular momentum l and magnetic quantum number m locate a label within its shell. On these labels the Laplace–Beltrami operator acts exactly as the diagonal $SO(4)$ Casimir $\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1$ (Eq. (7) below), and the continuous Schrödinger equation is its flat-space image under Fock's 1935 stereographic map [3, 5]. (The label lattice's graph Laplacian is *not* equivalent to that operator and does not converge to it: it preserves l and splits into disconnected l-blocks, Paper 7; corrected 2026-09-28 from "mathematically equivalent".)

This geometric picture is complete for single atoms. For molecules, the current implementation stitches together two separate S^3 manifolds—one per atom—using approximate cross terms: Mulliken overlap for cross-nuclear attraction, Fourier convolution for direct Coulomb integrals, and STO-overlap bridges for inter-center hopping [4]. The result is basis set superposition error (BSSE), approximate integrals, and no natural coordinate system for the bonding region.

This paper proposes the resolution: the *bond sphere*. A diatomic molecule is described by a single S^3 with two weighted poles, one per nucleus. The atomic S^3 manifolds are the $R \to \infty$ limit. Cross-atom matrix elements that are currently approximated become exact $SO(4)$ Wigner D-matrix elements. The natural basis for this geometry is the Sturmian basis, in which all functions share a single momentum parameter p_0 and therefore live on the same S^3 by construction. The β_k eigenvalues that encode nuclear attraction are determined by the two-center prolate spheroidal eigenvalue problem—whose coordinate surfaces are precisely the confocal ellipsoids of the bond sphere metric.

The paper is organized as follows. Sections III and IV establish the single-atom and diatomic bond sphere geometry. Section V gives the Runge–Lenz alignment argument for why the bond rotation is generated by A_y. Section VI treats heteronuclear asymmetry—the egg. Section VII identifies the kinetic membrane as the Fiedler vector zero-crossing. Section VIII introduces the Sturmian basis. Section IX develops the β_k eigenvalue problem via prolate spheroidal separation. Section X proves why atom-centered approximations fail. Section XI states the Sturmian diagonal theorem. Section XII proves the Sturmian structural theorem: in the single-n D-matrix approximation, the whole R-dependence of (H, S) is a single $SO(4)$ congruence that cancels in the generalized eigenvalue problem, making the eigenvalues geometry-independent (the genuine cross-n overlap escapes it, Corollary 2). Section XIII extends the structural theorem to multi-electron systems via Coulomb and generalized Sturmian CI. Section XIV derives the $SO(4)$ selection rules for σ-bond formation. Section XV concludes.

* Paper 8–9 in the Geometric Vacuum series

II. WHAT IS KNOWN VS. WHAT IS NEW

A. Known Results (Prior Work)

- *Shibuya and Wulfman (1965)* [6]: The two-center Coulomb problem in momentum space maps to a single S^3 with two poles. Each center's Sturmian functions on this sphere provide a complete expansion, orthonormal in the potential-weighted metric; the two-center set is not mutually orthogonal.

- *Aquilanti and Cavalli (1980s–1990s)* [7, 8]: Hyperspherical harmonic approach to molecular problems, including systematic use of $SO(4)$ Wigner D-matrices and the relationship between internuclear distance and angular pole separation on S^3.

- *Avery (2000s)* [10]: Extended the Sturmian basis to general molecular systems with practical computational methods. De Fazio, Cavalli, and Aquilanti [21] developed the *hyperquantization algorithm*, discretizing scattering problems on S^3 via hyperspherical harmonics—arriving at sparse discrete Hamiltonians from the continuous side, converging with GeoVac's label-first construction (see Sec. XIII). Avery [23] formulated the generalized Sturmian secular equation for multi-electron atoms using configuration-dependent scaling parameters.

- $SO(4)$ *representation theory* [11, 12]: The Wigner D-matrix elements of $SO(4)$ in the (n, l, m) basis, generated by angular momentum $\mathbf{L}$ and the Runge–Lenz vector $\mathbf{A}$.

- *Sturmian basis functions*: Established by Shull and Löwdin [13], Rotenberg [14], and Aquilanti–Avery [15, 16]. Orthogonality under $1/r$ weight and completeness for the Coulomb problem.

B. New Contributions

1. **Bond sphere as a GeoVac lattice object.** The bond sphere is the natural molecular generalization of the single-atom S^3 lattice (Papers 0, 1, 7), with within-block edges generated by $T_\pm, L_\pm$ and cross-block edges weighted by D-matrix elements at $g(\gamma(R))$.

2. **Runge–Lenz alignment argument** (Sec. V). The poles are forced onto the bond axis by dynamics: the Runge–Lenz vector of each atom orients toward the other nucleus.

3. **Heteronuclear egg geometry** (Sec. VI). For $Z_A \neq Z_B$, the bond sphere metric is a prolate spheroid with the heavier nucleus at the fat end.

4. **Kinetic membrane** (Sec. VII; **[UNVERIFIED]**, no backing test). The Fiedler vector zero-crossing of the graph Laplacian (not an $SO(4)$-covariant operator) is proposed as a combinatorial analog of Bader's bond critical point.

5. **Identification that the bond sphere requires the Sturmian basis** (Sec. VIII). The single-S^3 geometry is only consistent when all basis functions share one p_0.

6. **Proof that atom-centered Sturmian bases fail for heteronuclear molecules** (Sec. X), with the precise threshold $\beta > p_0/(2Z_B)$ for binding.

7. **Sturmian diagonal theorem** (Sec. XI). In the molecular Sturmian basis, the diagonal $\varepsilon_k = \frac{1}{2}p_0^2 - p_0^2/\beta_k$ is fixed by the Sturmian eigenvalue β_k alone, independent of (l, m) and γ.

8. **Sturmian structural theorem** (Sec. XII). In the molecular Sturmian basis with shared p_0, in the single-n D-matrix approximation, the one-electron (H, S) obey the two-term identity $H_{ij} = -\frac{1}{2}p_0^2 S_{ij} + (1 - \beta_j)V_{ij}$ and share a single $SO(4)$ congruence $D^{(n)}(\gamma(R))$, so the eigenvalues are independent of the bond angle γ and hence of R. The genuine cross-n Coulomb–Sturmian overlap escapes the theorem and binds, supplying the R-dependence algebraically through the Shibuya–Wulfman overlap (Corollary 2), but its non-orthogonality mixes l across centres and hits an l-mixing qubit-encoding wall (orthogonalization destroys the Gaunt l-sparsity).

9. $SO(4)$ **selection rules for σ-bonds** (Sec. XIV). $D^2_{10,10}(\gamma) = 1$ and $D^2_{00,10}(\gamma) = 0$ for all γ: the bond rotation preserves $m = 0$ within $l = 1$ and does not mix s and p states.

III. THE SINGLE-ATOM CASE AS A SPECIAL LIMIT

The existing GeoVac single-atom lattice is the $\gamma = 0$ limit of the bond sphere, so all results of Papers 0–7 are preserved.

A. The $T_\pm, L_\pm$ Operators as $SO(4)$ Generators

The GeoVac paraboloid lattice is generated by four transition operators (Paper 7, Eqs. (5)–(6)):

$$L_\pm|n, l, m\rangle \propto |n, l, m \pm 1\rangle, \tag{1}$$

$$T_\pm|n, l, m\rangle \propto |n \pm 1, l, m\rangle. \tag{2}$$

The angular momentum operators L_x, L_y, L_z satisfy $\mathfrak{su}(2)$:

$$[L_i, L_j] = i\epsilon_{ijk}L_k. \tag{3}$$

The scaled Runge–Lenz vector $\mathbf{A}$ acts within each n-shell, changing l by ± 1 (it is *not* generated by $T_\pm$, which change n; corrected 2026-09-28), and satisfies [11]:

$$[L_i, A_j] = i\epsilon_{ijk} A_k, \tag{4}$$

$$[A_i, A_j] = i\epsilon_{ijk} L_k. \tag{5}$$

The $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ structure is exhibited by

$$\mathbf{J}^{(\pm)} = \tfrac{1}{2}(\mathbf{L} \pm \mathbf{A}), \tag{6}$$

which satisfy two independent, commuting $\mathfrak{su}(2)$ algebras. For principal quantum number n, both carry spin $j = (n-1)/2$, giving the $(n-1)/2 \otimes (n-1)/2$ representation with Casimir

$$\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1, \tag{7}$$

recovering the S^3 eigenvalue spectrum $\lambda_n = -(n^2-1)$ of Paper 7.

B. Covariance and the Single-Atom Limit

Because $\mathbf{L}$ and $\mathbf{A}$ generate $SO(4)$ within each n-shell (Eqs. (4)–(5)), each shell of the label set carries the irreducible representation of Eq. (7), and the Casimir $\mathbf{L}^2+\mathbf{A}^2$ is $SO(4)$-invariant on the labels. (The lattice's graph Laplacian $L = D - A_{\text{adj}}$, with A_{adj} the adjacency matrix and not the Runge–Lenz vector $\mathbf{A}$, is *not* $SO(4)$-covariant: it preserves l, while $\mathbf{A}$ mixes l within a shell. Nothing in the structural theorem (Sec. XII) uses the graph Laplacian—the kinetic-membrane construction of Sec. VII does; the rotated-copy construction acts on the label set and the Sturmian functions it indexes.) The second atom can be introduced as an $SO(4)$-rotated copy of the label set, with the rotation angle determined by R. Cross-atom matrix elements are then $SO(4)$ Wigner D-matrix elements.

At $\gamma = 0$ (coincident poles): $g(0) = e$, the D-matrix reduces to Kronecker deltas, cross-block edges vanish, and the single-center lattice of Papers 0–7 is recovered. The 18 symbolic checks, the dimensionless vacuum principle, and the kinetic scale factor $-1/16$ are unchanged.

IV. THE BOND SPHERE CONSTRUCTION

A. The Angle–Distance Relationship

The internuclear distance R maps to a polar angle γ between two poles on S^3. The momentum conjugate to R defines

$$p_R = \frac{1}{R}, \tag{8}$$

and the angular separation is

$$\cos\gamma = \frac{p_0^2 - p_R^2}{p_0^2 + p_R^2}, \tag{9}$$

which is the fourth component n_4 of the Fock projection evaluated at $|\mathbf{p}| = p_R$. The sine follows as

$$\sin\gamma = \frac{2p_0\, p_R}{p_0^2 + p_R^2} = \Omega(p_R) \cdot p_R, \tag{10}$$

where Ω is the Fock conformal factor.

a. Limiting behavior. At $R \to \infty$: $\gamma \to 0$ (coincident poles, single-atom limit). At $R \to 0$: $\gamma \to \pi$ (antipodal poles, united-atom limit).

B. Eigenfunctions as $SO(4)$ Wigner D-Matrix Elements

When a second pole is introduced at angle γ, the cross-center matrix elements are

$$\langle n', l', m' | U(g(\gamma)) | n, l, m \rangle = D^{(n)}_{(l'm'),(lm)}(g(\gamma)), \tag{11}$$

where $g(\gamma)$ is the $SO(4)$ rotation that maps one pole to the other. The current Mulliken/Fourier/bridge cross-atom terms are approximations to these exact D-matrix elements.

C. The Bond-Sphere Lattice

a. Nodes. Each node is labeled (n, l, m, α) with $\alpha \in \{A, B\}$. Total spatial orbitals: $2N_{\text{sp}}$.

b. Within-block edges. Generated by $T_\pm, L_\pm$ at each center, as in the single-atom lattice.

c. Cross-block edges. Weighted by D-matrix elements:

$$w_{(n'l'm',B),(nlm,A)} = D^{(n)}_{(l'm'),(lm)}(g(\gamma(R))). \tag{12}$$

d. Representation-theoretic constraint. The D-matrix $D^{(n)}_{(l'm'),(lm)}(\gamma)$ is block-diagonal in the principal quantum number n by the representation theory of $SO(4)$, not by approximation. Each n-shell corresponds to a distinct irreducible representation of $SO(4)$ with $j = (n-1)/2$; different irreps are orthogonal under any group element. The molecular field (second nucleus at bond angle γ) acts as an $SO(4)$ rotation, which cannot mix inequivalent representations.

Cross-n coupling—for example, Li $2s \leftrightarrow$ H $1s$ in heteronuclear molecules—is generated by placing the second Coulomb centre at distance R, which is a *translation* on the Fock sphere, not an $SO(4)$ rotation. A translation expands over hyperspherical harmonics of every degree $\lambda = n - 1$ and therefore couples different n-shells; it is captured in closed form by the many-centre Coulomb–Sturmian / Shibuya–Wulfman matrix, a function of the dimensionless product $s = p_0 R$ (not a discrete rotation angle), for which explicit, programmable formulae exist in the literature [24, 25]. The block-diagonal $D^{(n)}(\gamma)$ used in the bond-sphere lattice below is thus a *single-n approximation* of that exact cross-n object, adopted

here for the discrete-graph construction; its known cost is exactly the missing cross-n σ hybridization noted next. (Earlier drafts asserted that no closed-form cross-nuclear coupling existed and that the Shibuya–Wulfman formula was single-n only; both statements were incorrect—the SW matrix *is* the cross-n expansion—and are retracted here.)

This block-diagonality is the fundamental reason why LCAO approaches using D-matrix coupling fail to capture σ hybridization between orbitals of different principal quantum number. The symmetry breaking $SO(4) \to SO(2)$ caused by the second nucleus (Sec. VI) breaks l-degeneracy *within* each n-shell but does *not* couple different n-shells.

e. Hamiltonian structure.

$$H_1 = \begin{pmatrix} H_A & V_{AB} \\ V_{AB}^\dagger & H_B \end{pmatrix}, \tag{13}$$

where H_A, H_B are single-atom Hamiltonians and V_{AB} has elements given by Eq. (12) modulated by nuclear charges and kinetic scale.

D. The Opposite-Sign Rotation

The bond rotation is not a spatial rotation. Since $\mathbf{A} = \mathbf{J}^{(+)} - \mathbf{J}^{(-)}$, the rotation generated by A_y acts as

$$e^{i\gamma A_y} = e^{i\gamma J_y^{(+)}}\, e^{-i\gamma J_y^{(-)}}, \tag{14}$$

where the $\mathbf{J}^{(+)}$ and $\mathbf{J}^{(-)}$ sectors rotate in opposite directions: $+\gamma$ and $-\gamma$. A pure spatial rotation $e^{i\alpha L_y}$ gives same-sign rotation $d^j(+\alpha) \otimes d^j(+\alpha)$, preserving radial structure. The bond rotation gives $d^j(+\gamma) \otimes d^j(-\gamma)$, mixing $s, p, d, \ldots$ orbitals within each n-shell. This s–p hybridization is precisely what chemical bonding requires.

The D-matrix in the (n, l, m) basis is

$$D^{(n)}_{(l'm'),(lm)}(\gamma) = \sum_{m^+,m^-} \sum_{m'^+,m'^-} C^n_{l'm';m'^+m'^-}\, C^n_{lm;m^+m^-} \times\, d^j_{m'^+,m^+}(+\gamma)\, d^j_{m'^-,m^-}(-\gamma), \tag{15}$$

where $j = (n-1)/2$ and $C^n_{lm;m^+m^-}$ are the $SO(4)$ Clebsch–Gordan coefficients coupling $|j, m^+\rangle \otimes |j, m^-\rangle \to |l, m\rangle$ with $m = m^+ + m^-$.

E. The Antipodal Limit

At $\gamma = \pi$ ($R = 0$, united atom):

$$D^{(n)}(\pi)^2 = \mathbf{I}_{n^2}, \tag{16}$$

with eigenvalues ± 1 and parity structure $|l, m\rangle \to (-1)^{n-1+l+|m|}\, |l, -m\rangle$—the Runge–Lenz parity, not spatial parity $(-1)^l$.

V. THE RUNGE–LENZ ALIGNMENT ARGUMENT

The bond rotation $e^{i\gamma A_y}$ is generated by A_y, a component of the Runge–Lenz vector perpendicular to the bond axis. This choice is not a convention but a dynamical consequence.

For a hydrogen atom, the Runge–Lenz vector $\mathbf{A}$ points from the nucleus toward the perihelion of the classical orbit. In a diatomic molecule, the electrostatic field of the partner nucleus breaks the spherical symmetry of each atom's potential. The lowest-energy configuration aligns the Keplerian orbit so that its perihelion points toward the attracting partner—that is, $\mathbf{A}$ orients along the internuclear axis. With $\mathbf{A}$ aligned along $\hat{z}$ (the bond axis), the rotation that maps one pole to the other is generated by A_y (or A_x), which is perpendicular to the bond.

This is the physical origin of s–p hybridization in the bond sphere picture: the Runge–Lenz vector aligns with the bond, and rotation about a perpendicular axis mixes orbital angular momentum states. The hybridization is not an ansatz imposed on the basis—it is a consequence of the two-center electrostatic potential acting on the dynamical symmetry generators of each atom.

VI. HETERONUCLEAR GEOMETRY: THE EGG

For a homonuclear diatomic ($Z_A = Z_B$), the bond sphere has uniform metric and full $SO(4)$ symmetry. For a heteronuclear diatomic ($Z_A \neq Z_B$), the geometry changes qualitatively.

A. Metric Distortion

The Fock conformal factor $\Omega = 2p_0/(p^2 + p_0^2)$ maps momentum space to S^3. When the two nuclei have unequal charges, the natural momentum scale differs between the two poles: center A (charge Z_A) favors $p_0 \sim Z_A$, while center B (charge Z_B) favors $p_0 \sim Z_B$. With a shared p_0 (as required by the single-S^3 construction), the conformal factor is distorted asymmetrically about the two poles.

The result is a bond sphere whose intrinsic metric is not uniform. The heavier nucleus ($Z_A > Z_B$) occupies a larger solid angle on the sphere—a "fat end"—while the lighter nucleus occupies a smaller solid angle—a "narrow end." The overall shape is a prolate spheroid, with the heavier nucleus at the fat end and the lighter nucleus at the narrow end. We call this the *egg*.

B. Bond Polarity as Egg Asymmetry

The degree of asymmetry is set by the ratio Z_A/Z_B. For LiH ($Z_A/Z_B = 3$), the egg is noticeably asymmetric.

For H_2 ($Z_A/Z_B = 1$), the egg is a round sphere. This asymmetry is the geometric encoding of bond polarity: a polar bond is an asymmetric egg, and a nonpolar bond is a symmetric sphere.

C. Topology vs. Metric

The node count per shell (n^2 nodes at latitude n from each pole) is fixed by the Paper 0 packing topology. The egg deformation changes the *node weights*—the effective nuclear attraction felt by each node—but not the *graph topology*. The adjacency structure ($T_\pm, L_\pm$ edges) is unchanged; only the diagonal elements of the Hamiltonian (the node weights) and the cross-block edge weights (the D-matrix elements) are affected by the asymmetry.

D. Symmetry Breaking: $SO(4) \to SO(2)$

The unequal pole weights break $SO(4)$ to the axial $SO(2)$ subgroup preserving the bond axis. The magnetic quantum number m remains a good quantum number (generating $\sigma, \pi, \delta, \ldots$ labels), while the off-axis generators $L_\pm$ are no longer symmetries:

$$[\Delta_{\text{asym}}, L_z] = 0, \tag{17}$$

$$[\Delta_{\text{asym}}, L_\pm] \neq 0 \quad \text{for } Z_A \neq Z_B. \tag{18}$$

E. Prolate Spheroidal Coordinates

The natural coordinate system for the egg is prolate spheroidal coordinates (ξ, η, φ) with foci at the two nuclear poles. The coordinate surfaces are confocal ellipsoids ($\xi = \text{const}$) and hyperboloids ($\eta = \text{const}$). These are precisely the level sets of the distorted bond sphere metric: the ellipsoids trace curves of constant "distance from both poles simultaneously," while the hyperboloids trace curves of constant "distance ratio between the two poles." This coordinate system appears again in Sec. IX as the natural domain for the β_k eigenvalue problem.

VII. THE KINETIC MEMBRANE

Between the two poles of the bond sphere lies a region where the edge weights from the two centers are comparable in magnitude. This is the *kinetic membrane*—not a separate geometric object, but an emergent feature of the edge weight pattern.

A. Identification via the Fiedler Vector

In the molecular graph, the kinetic membrane is identified by the *Fiedler vector*—the eigenvector corresponding to the second smallest eigenvalue of the graph Laplacian. The Fiedler vector partitions the graph into two regions (positive and negative components), and the membrane is where the Fiedler vector crosses zero.

For a homonuclear diatomic, the Fiedler vector zero-crossing lies at the midpoint between the two poles, by symmetry. For a heteronuclear molecule, the zero-crossing is expected to shift toward the lighter nucleus: in LiH, the heavier Li pole ($Z = 3$) dominates a larger volume of the bond sphere, pushing the membrane toward the H end. **[UNVERIFIED]**: no test computes this shift.

B. Connection to Bader's Bond Critical Point

The Fiedler vector zero-crossing is the discrete, combinatorial analog of Bader's bond critical point [17] in the Quantum Theory of Atoms in Molecules (QTAIM). Bader's bond critical point is defined as the point where $\nabla\rho = 0$ along the bond path, where ρ is the electron density. The GeoVac graph gives a discrete analog of this object without invoking the electron density: the membrane position is determined purely by the graph Laplacian spectrum. The analogy is at the level of graph partitioning only: since the graph Laplacian does not carry the one-electron spectrum (it does not converge to Δ_{S^3}, Paper 7), the membrane is a combinatorial feature of the chosen edge weights, not a computed property of the electron density.

For heteronuclear molecules the Bader critical point shifts toward the electropositive atom — in LiH that is Li, the *heavier* atom — whereas the Fiedler zero-crossing described above shifts toward the lighter nucleus (H in LiH). For LiH the two therefore move in *opposite* directions, so the analogy does not extend to the direction of the shift (corrected 2026-09-28: this sentence previously said both shift toward "the electropositive (lighter) atom"). The statement below is likewise **[UNVERIFIED]**. The membrane is thinner (steeper Fiedler gradient) for strongly polar bonds and thicker (shallower gradient) for nonpolar bonds.

VIII. THE STURMIAN BASIS

The bond sphere hypothesis requires all basis functions to live on a single S^3 parameterized by one momentum scale p_0. The hydrogenic eigenstate basis uses $p_0(n) = Z/n$—a different sphere per shell—which is geometrically inconsistent with the single-S^3 construction. The Sturmian basis resolves this.

A. Definition

The Sturmian basis functions S_{nlm} are solutions to the scaled Coulomb problem:

$$\left(-\tfrac{1}{2}\nabla^2 - \frac{\beta_n Z}{r}\right) S_{nlm} = E_0\, S_{nlm}, \tag{19}$$

where $E_0 = -p_0^2/2$ is a fixed reference energy shared by all functions, and

$$\beta_n = \frac{p_0\, n}{Z} \tag{20}$$

is the shell-dependent scaling that enforces this.

The key property: all Sturmian functions share the same E_0 regardless of n. Under Fock's projection, all functions live on the *same* S^3 of radius p_0.

B. Verification

Substituting effective charge $\tilde{Z} = \beta_n Z$ into the hydrogenic eigenvalue:

$$E_0 = -\frac{(\beta_n Z)^2}{2n^2} = -\frac{(p_0 n/Z)^2 Z^2}{2n^2} = -\frac{p_0^2}{2}, \tag{21}$$

independent of n, as required.

C. Orthogonality under $1/r$ Weight

Sturmian functions satisfy

$$\left\langle S_{n'l'm'} \left| \frac{1}{r} \right| S_{nlm} \right\rangle = \frac{p_0}{n}\, \delta_{nn'}\, \delta_{ll'}\, \delta_{mm'}. \tag{22}$$

The $1/r$-weighted completeness precisely matches the Coulomb kernel in cross-center integrals, making the two-center problem exact.

D. Geometric Inconsistency of the Hydrogenic Basis

The hydrogenic eigenstate ψ_{nlm} has $p_0(n) = Z/n$. For the cross-center matrix element $\langle \psi^A_{n'l'm'} | -Z_B/r_B | \psi^A_{nlm} \rangle$ with $n' \neq n$, the bra lives on S^3 with radius Z/n' and the ket on S^3 with radius Z/n. The integral involves two different Fock projections—a convolution over mismatched spheres. This is the origin of the $\sin\gamma$ form factor in the current SW module: a lowest-order correction for this geometric mismatch.

In the Sturmian basis, the mismatch is zero by construction.

E. SW Integrals Are Exact in the Sturmian Basis

This is the central result. In the Sturmian basis:

$$\boxed{\left\langle S^A_{n'l'm'} \left| -\frac{Z_B}{r_B} \right| S^A_{nlm} \right\rangle = -\frac{Z_B}{p_0}\, D^{(n)}_{(l'm'),(lm)}(\gamma),} \tag{23}$$

exactly, with no form factor.

The derivation: the cross-center Coulomb potential is related to the same-center potential by an $SO(4)$ rotation (Eq. (11)). Using Sturmian orthogonality (Eq. (22)), the $1/r_A$ matrix element collapses to a Kronecker delta, and the sum over intermediate states reduces to a single D-matrix element.

When the same calculation is attempted with hydrogenic eigenstates of different p_0 values, the orthogonality does not apply, and one obtains the mismatch integral that, expanded to leading order, gives the $\sin\gamma$ form factor.

F. The Self-Consistency Condition

The momentum parameter p_0 is determined by

$$p_0 = \sqrt{-2E_{\mathrm{mol}}}, \tag{24}$$

where E_{mol} is the ground-state eigenvalue of $H(p_0)$. This is a self-consistent fixed point, whose existence follows from the intermediate value theorem: $f(p_0) = \sqrt{-2E_{\mathrm{mol}}(p_0)}$ maps $\mathbb{R}^+$ to $\mathbb{R}^+$, with $f(p_0) > p_0$ for small p_0 (diffuse basis, weakly negative energy) and $f(p_0) < p_0$ for large p_0 (compact basis, bounded energy).

G. γ Encodes Both Geometry and Energy

At self-consistency:

$$\gamma = \arccos\left(\frac{p_0^2 - p_R^2}{p_0^2 + p_R^2}\right), \tag{25}$$

with $p_0 = \sqrt{-2E_{\mathrm{mol}}}$. Both derivatives $\partial\gamma/\partial R \neq 0$ and $\partial\gamma/\partial E_{\mathrm{mol}} \neq 0$, confirming that γ carries information about both bond length and molecular energy simultaneously.

H. Backward Compatibility with Papers 0–7

In the limit $p_0 \to Z/n$ per shell: $\beta_n \to 1$, the Sturmian equation becomes the ordinary hydrogenic equation, and the atomic eigenvalue $-Z^2/(2n^2)$ is recovered. The self-consistency loop has no single-atom analog because $E_n = -Z^2/(2n^2)$ already determines $p_0(n) = Z/n$. All single-atom results are the n-resolved limit of the Sturmian construction.

IX. THE β_k EIGENVALUE PROBLEM

For a single-center Sturmian, $\beta_n = p_0 n/Z$ is determined by the nuclear charge alone. For a two-center molecular Sturmian, β_k is the eigenvalue of the prolate spheroidal equation with foci at the two nuclei.

A. Prolate Spheroidal Separation

The Sturmian equation with the two-center molecular potential $V = -Z_A/r_A - Z_B/r_B$ separates in prolate spheroidal coordinates (ξ, η, φ) with foci at the nuclear positions. The separation yields an angular eigenvalue problem (in η) and a radial eigenvalue problem (in ξ), coupled by a separation constant A.

The angular equation at fixed magnetic quantum number m and spheroidal parameter $c = p_0 R/2$ is

$$\frac{d}{d\eta}\left[(1-\eta^2)\frac{dS}{d\eta}\right] + \left(A_{m,n_{\rm sph}} - c^2\eta^2 - \frac{m^2}{1-\eta^2}\right) S = 0, \tag{26}$$

whose eigenvalues $A_{m,n_{\rm sph}}(c, b)$ depend on c and the charge asymmetry parameter $b = (Z_A - Z_B)R/2$.

The radial equation is

$$\frac{d}{d\xi}\left[(\xi^2-1)\frac{dR}{d\xi}\right] + \left(-A + c^2\xi^2 + 2a\xi - \frac{m^2}{\xi^2-1}\right) R = 0, \tag{27}$$

where $a = \beta_k(Z_A + Z_B)R/2$ encodes the molecular Sturmian eigenvalue β_k.

B. The Matching Condition

The molecular β_k for each orbital $(m, n_{\rm sph}, n_{\rm rad})$ is determined by the matching condition: the angular separation constant $A_{m,n_{\rm sph}}(c, b)$ and the radial eigenvalue must be consistent. Specifically, β_k is found by root-finding (brentq) on the condition

$$A_{\rm ang}(c, b) + L_{\rm rad}(c, a(\beta_k)) = 0, \tag{28}$$

where $L_{\rm rad}$ is the top radial eigenvalue at the given parameters.

C. Key Result

Theorem 1 (Molecular Sturmian β_k for LiH)**.** *For LiH at $R = 3.015$ bohr with $p_0 = \sqrt{-2E_{mol}}$, the molecular Sturmian β_k values computed via prolate spheroidal separation are:*

Orbital	β_k	*Character*
1σ	*1.033*	*Li 1s core*
2σ	*1.950*	*Li-H bonding*
1π	*1.976*	*Li-H π*

All $n = 3$ orbitals have $\beta_k > 2.3$. The H_2^+ ground-state energy computed by this method is $E = -1.096$ Ha (exact: -1.103 Ha, 0.6% error).

D. Why Prolate Spheroidal Coordinates Are Natural

The separability of the Sturmian equation in prolate spheroidal coordinates is not a mathematical accident. The coordinate foci are the two nuclear poles, and the coordinate surfaces are the confocal ellipsoids that are the level sets of the distorted bond sphere metric (Sec. VI). The coordinates respect both poles simultaneously—which is precisely the requirement for a basis that lives on the bond sphere rather than on a single-center sphere.

X. WHY ATOM-CENTERED BASES FAIL FOR HETERONUCLEAR MOLECULES

Theorem 2 (Atom-centered Sturmian failure)**.** *For a heteronuclear diatomic with $Z_A/Z_B = k > 1$, there is no atom-centered Sturmian basis with shared p_0 that simultaneously binds both centers.*

Proof sketch. The shared p_0 is determined by the deeper well (Z_A). At this p_0, the one-electron diagonal for the lightest orbital of center B is

$$\varepsilon_1(p_0) = \frac{p_0^2}{2} - Z_B\, p_0. \tag{29}$$

This is positive (unbound) when $p_0 > 2Z_B$. For the self-consistent $p_0 \approx \sqrt{-2E_A}$ of the heavier atom with $Z_A \gg Z_B$, this condition is generically satisfied.

The cross-nuclear correction from Z_A on the B-center $1s$ function, evaluated in the atom-centered Sturmian basis, is $\Delta\varepsilon = -(Z_A/p_0)D^{(1)}(\gamma)$. For $n = 1$, $D^{(1)} = 1$, so $\Delta\varepsilon = -Z_A/p_0$. The total diagonal becomes

$$\varepsilon_1^{\rm tot} = \frac{p_0^2}{2} - Z_B p_0 - \frac{Z_A}{p_0}. \tag{30}$$

For binding: $\varepsilon_1^{\rm tot} < 0$, i.e., $\beta_{\rm eff} > p_0/(2Z_B)$. But the atom-centered $\beta_1 = p_0/Z_B$ gives $\varepsilon_1 = p_0^2/2 - Z_B p_0$ plus the cross-nuclear $-Z_A/p_0$. At $p_0 = 3.168$ (LiH): $\varepsilon_1^{\rm tot} = 5.02 - 3.17 - 0.95 = +0.85$ Ha—still positive.

The cross-nuclear correction is insufficient because the atom-centered function has the wrong spatial extent: it was defined against Z_B alone and is too compact near the B center to capture the full A-B cross-nuclear attraction. The correct integral requires the *molecular* Sturmian function—the two-center prolate spheroidal eigenfunction—which has the correct spatial extent near both centers. □

Corollary 1. *For LiH ($Z_A/Z_B = 3$): the binding threshold is $\beta > p_0/(2Z_B) = 1.581$. The 2σ molecular Sturmian achieves $\beta = 1.950$ (Theorem 1), but no atom-centered approximation reaches this threshold.*

This argument is independent of the number of electrons and applies to any heteronuclear diatomic with Z_A/Z_B sufficiently different from 1. The diagnostic arc v0.9.20–v0.9.30 (documented in the companion paper [4]) exhaustively verifies this theorem computationally: every atom-centered variant fails for LiH by the mechanism described above.

XI. THE STURMIAN DIAGONAL THEOREM

Theorem 3 (Sturmian diagonal)**.** *In the molecular Sturmian basis with shared p_0, the diagonal of the one-electron Hamiltonian is fixed by the Sturmian eigenvalue β_k alone:*

$$\varepsilon_k = -\frac{p_0^2}{2} + (1-\beta_k)\,\langle S_k|V_{\rm mol}|S_k\rangle = \frac{p_0^2}{2} - \frac{p_0^2}{\beta_k}, \qquad (31)$$

independent of the angular quantum numbers (l,m) and of the bond geometry γ. For atomic Coulomb Sturmians, where β depends only on the shell ($\beta_n = p_0 n/Z$), ε_k is uniform within each n-shell. (It equals $-p_0^2/2$ only in the degenerate case $\beta_k = 1$, where the Sturmian coincides with a physical eigenstate.)

Proof. The molecular Sturmian S_k satisfies $(-\frac{1}{2}\nabla^2 + \beta_k V_{\rm mol})S_k = E_0 S_k$ with $E_0 = -p_0^2/2$. Writing the full one-electron Hamiltonian as $\hat{h}_1 = (-\frac{1}{2}\nabla^2 + \beta_k V_{\rm mol}) + (1-\beta_k)V_{\rm mol}$ and taking the (normalized) diagonal expectation value gives the first equality,

$$\varepsilon_k = E_0 + (1-\beta_k)\,\langle S_k|V_{\rm mol}|S_k\rangle. \qquad (32)$$

The virial theorem for the homogeneous Coulomb potential $\beta_k V_{\rm mol} \propto 1/r$ gives $2\langle T\rangle = -\beta_k\langle V_{\rm mol}\rangle$; combined with $\langle T\rangle + \beta_k\langle V_{\rm mol}\rangle = E_0 = -p_0^2/2$ this fixes $\langle S_k|V_{\rm mol}|S_k\rangle = -p_0^2/\beta_k$, giving the second equality

$$\varepsilon_k = -\frac{p_0^2}{2} + (1-\beta_k)\left(-\frac{p_0^2}{\beta_k}\right) = \frac{p_0^2}{2} - \frac{p_0^2}{\beta_k}. \qquad (33)$$

□

a. Physical interpretation. The diagonal energy is fixed entirely by the Sturmian eigenvalue β_k; it is blind to the angular quantum numbers (l,m) and to the bond angle γ. All geometric and angular differentiation between orbitals is therefore carried by the *off-diagonal* structure—the D-matrix coupling weighted by β_k-dependent factors—rather than by the orbital energies. The asymmetry between the two centers enters only through the β_k eigenvalues (which differ for σ vs. π MOs and depend on Z_A, Z_B, R) and through the D-matrix rotation angle γ.

b. Consistency with the β_k discussion. The β_k eigenvalues (Sec. IX) parametrize the nuclear attraction: orbital k has effective charge $\beta_k Z$ in the Sturmian equation. They set the diagonal energies through $\varepsilon_k = \frac{p_0^2}{2} - p_0^2/\beta_k$ (Eq. (31)) and weight the off-diagonal D-matrix coupling. What is *absent* from the diagonal is the angular and geometric content: ε_k records only the radial Sturmian scale β_k and is independent of (l,m) and of $\gamma(R)$, so all bond-geometry dependence lives off-diagonal. This is the correct architecture for a bond sphere FCI: the basis is adapted to the molecular potential, the diagonal records only the radial Sturmian scale, and the bond geometry appears in the coupling structure.

XII. GEOMETRIC LIMITS OF THE STURMIAN BOND SPHERE

The computational diagnostic arc v0.9.20–v0.9.34 (29 versions) exhaustively tested the Sturmian bond sphere for LiH. Every variant failed. This section distills the failure into a single structural theorem that explains why the Sturmian basis, despite its geometric elegance, cannot produce geometry-dependent binding in this single-n approximation without solving a continuous PDE at every R—while the genuine cross-n Coulomb–Sturmian overlap escapes the theorem and binds (Remark 1), so the obstruction is to the approximation, not to Sturmian molecular binding.

Theorem 4 (Sturmian structural theorem)**.** *In the molecular Sturmian basis with shared momentum scale p_0, the one-electron Hamiltonian and overlap matrices obey the two-term Sturmian identity*

$$H_{ij} = -\frac{p_0^2}{2}\,S_{ij} + (1-\beta_j)\,V_{ij}, \qquad (34)$$

where $S_{ij} = \langle S_i|S_j\rangle$ is the overlap matrix, $V_{ij} = \langle S_i|V_{\rm mol}|S_j\rangle$ is the nuclear-attraction matrix, and β_j is the Sturmian eigenvalue (effective-charge multiplier, $\beta_n = p_0 n/Z$) of orbital j. The Hamiltonian is not *a scalar multiple of S; rather, in the bond-sphere construction both S and V (hence H) carry their* entire R*-dependence through the same $SO(4)$ Wigner D-matrix $D^{(n)}(\gamma(R))$: within-center blocks are R-independent (orthonormal Sturmians, $S = \mathbb{1}$; uniform self-attraction, $V = -\frac{p_0^2}{\beta}\mathbb{1}$, Theorem 3), while cross-center blocks are proportional to $D^{(n)}(\gamma)$. Consequently (H,S) at any two distances are related by the orthogonal congruence $U(\gamma) = \mathrm{blockdiag}(\mathbb{1}, D^{(n)}(\gamma))$, under which both matrices pull back to R-independent forms, so the generalized eigenvalues*

$$H\,\mathbf{c}_k = E_k\,S\,\mathbf{c}_k \qquad (35)$$

are independent of the bond angle γ and hence independent of the internuclear distance R.

Proof. Each Sturmian function S_j satisfies $(-\frac{1}{2}\nabla^2 + \beta_j V_{\rm mol})S_j = E_0 S_j$ with $E_0 = -p_0^2/2$ and $V_{\rm mol} = -Z_A/r_A - Z_B/r_B$. Writing the physical one-electron Hamiltonian as $\hat{h}_1 = -\frac{1}{2}\nabla^2 + V_{\rm mol} = (-\frac{1}{2}\nabla^2 + \beta_j V_{\rm mol}) +$

$(1-\beta_j)V_{\rm mol}$ and acting on S_j gives $\hat{h}_1 S_j = E_0 S_j + (1-\beta_j)V_{\rm mol}S_j$; projecting on $\langle S_i|$ yields Eq. (34). (The physical matrix $H_{ij} = \langle S_i|\hat{h}_1|S_j\rangle$ is Hermitian by self-adjointness of $\hat{h}_1$; for orbitals on the same center, or for a homonuclear shell with a single β, the column factor $(1-\beta_j)$ and the row factor $(1-\beta_i)$ already agree, and the production assembly symmetrizes the per-column form in the general heteronuclear case.)

Because $D^{(n)}(\gamma)$ is orthogonal, $D^{(n)}(\gamma)^\top D^{(n)}(\gamma) = \mathbb{1}$, the congruence $U(\gamma) = \mathrm{blockdiag}(\mathbb{1}, D^{(n)}(\gamma))$ satisfies $U^\top S(\gamma) U = S_0$ and $U^\top H(\gamma) U = H_0$ with S_0, H_0 independent of γ. Since $\det(H - ES) = \det(U)^2 \det(H_0 - ES_0)$, the generalized eigenvalues of (H, S) equal those of the R-independent pair (H_0, S_0) and are thus independent of γ, hence of R. The nuclear repulsion $V_{NN} = Z_A Z_B/R$ then adds a purely $1/R$ term, producing a monotonically repulsive total energy with no minimum. This is verified to machine precision by the bond-sphere backing test `tests/test_paper8_sturmian_structural.py`: across $R \in \{1.5, 3, 5, 8\}$ bohr the $D^{(n)}(\gamma(R))$ blocks vary strongly (spread > 0.3) yet the generalized eigenvalues of (H, S) are bit-exactly R-independent, and $E_{\rm total}(R)$ is a strictly decreasing pure-$1/R$ curve with no interior minimum (no bound state). □

Remark 1 (Scope: the overlap premise is imposed, not generic). The congruence hinges on *both* cross-blocks being the orthogonal $D^{(n)}(\gamma)$. For the cross-nuclear *potential* $\langle S^A|-Z_B/r_B|S^A\rangle$ this holds exactly (Eq. (23): the $1/r$ weight invokes Sturmian orthogonality, collapsing the sum to a single D-matrix element). The cross-center *overlap* $\langle S^A_{nlm}|S^B_{n'l'm'}\rangle$ carries no such $1/r$ weight and does *not* collapse: as a translation on the Fock sphere it couples different n (Sec. VI above and the SW literature [24, 25]). A direct evaluation confirms it—for shared p_0, e.g. $\langle\chi^A_{1s}|\chi^B_{2s}\rangle = -0.47$ and $\langle\chi^A_{1s}|\chi^B_{2p_0}\rangle = -0.52$ at $R = 3.015$, both larger than the within-n diagonal $\langle\chi^A_{1s}|\chi^B_{1s}\rangle = +0.35$. The backing test therefore *imposes* $S_{AB} = f\, D^{(n)}$ by construction rather than computing the genuine overlap; the geometry-independence it verifies is a property of the bond-sphere D-matrix *model*, not of the genuine cross-n Coulomb–Sturmian basis. Because a cross-n overlap breaks the shared congruence (S and H no longer pull back to the same R-independent forms), binding is *not* excluded for the genuine basis—consistent with published Coulomb–Sturmian Hartree–Fock calculations [26] (there for atoms; the molecular cross-n binding is the escape this Remark establishes). The theorem thus delimits a specific approximation; it is not a no-go for Sturmian molecular binding. (Backing: `tests/test_paper8_overlap_cross_n.py`.)

Corollary 2 (Binding requires R-dependent coupling)**.** *Within the single-n D-matrix approximation, binding requires an R-dependent spectrum $\beta_k(R)$ that the orthogonal congruence cannot supply. In the genuine cross-n basis this R-dependence is supplied* algebraically *by the Shibuya–Wulfman overlap, whose matrix elements are functions of the dimensionless product $s = p_0 R$ (Remark 1); it need not come from solving a continuous two-center PDE. The earlier statement that binding necessarily reintroduces continuous spatial geometry held only for the block-diagonal approximation and is superseded by the cross-n evaluation.*

Corollary 3 (Heteronuclear dual-p_0 failure)**.** *For heteronuclear diatomics, no shared p_0 exists that simultaneously represents both atomic length scales. The dual-p_0 extension (v0.9.34) fails because molecular Sturmian functions are inherently delocalized over both centers: at $p_{0,H} = 1.0$, the H-scale $1\sigma'$ orbital "sees" the Li nucleus ($Z_A = 3$) and acquires $H_1 = -2.6$ Ha. The self-consistent loop drives both p_0 values toward a single geometric mean ($p_0 \approx 2.0$), eliminating the scale separation that was the motivation for the extension.*

a. Sturmian basis for heteronuclear systems. The Sturmian basis resolves the geometric mismatch between n-shells by placing all orbitals on a single S^3 with shared momentum scale p_0. In this basis the cross-nuclear potential $-Z_B/r_B$ reduces exactly to a single D-matrix element (Eq. (23)), while different Sturmian n-shells genuinely couple through the Shibuya–Wulfman cross-center *overlap* (Remark 1)—in contrast to the hydrogenic basis, where the block-diagonality constraint forbids cross-n coupling.

However, the Sturmian approach fails for heteronuclear molecules (e.g., LiH) because no single p_0 simultaneously satisfies the self-consistency condition $p_0 = \sqrt{-2E}$ for both atoms (Corollary 3). Twenty-nine diagnostic versions (v0.9.20–34) explored Sturmian-based molecular Hamiltonians; all exhibited pathological behavior for LiH due to this p_0 incompatibility. The heteronuclear problem thus remains open.

XIII. MULTI-ELECTRON STURMIAN CI: COMPUTATIONAL VERIFICATION

The structural theorem (Theorem 4) proves that, in the single-n D-matrix approximation, the one-electron (H, S) share a single $SO(4)$ congruence—so the generalized eigenvalues are R-independent—for molecular Sturmians with shared momentum scale p_0. The natural question is whether multi-electron Sturmians—where the electron–electron repulsion V_{ee} introduces coupling beyond one-body structure—can circumvent this limitation and produce basis sets that outperform standard hydrogenic FCI for atoms and molecules. Tracks BU-1 and BU-1b (v2.0.33) tested this for helium (two electrons) using both Coulomb and generalized Sturmian CI.

a. Convergent discovery: Aquilanti and Avery. The GeoVac framework and the Aquilanti group (Perugia) [7–9] independently arrived at the same mathematical structure—Fock projection onto S^3, hyperspherical harmonic expansions, angular momentum algebra via Wigner $3j$ and Gaunt coefficients, and sparse discrete

Hamiltonians—from opposite directions. GeoVac proceeds labels → continuous (Paper 7 [3]): the discrete label lattice is its starting object (a choice of direction, not a claim of priority), and the Schrödinger equation is reached via stereographic projection of the S^3 operator acting on those labels (as the diagonal $SO(4)$ Casimir; the lattice's graph Laplacian does not converge to it). Aquilanti and collaborators proceed continuous → discrete: the hyperquantization algorithm [21, 22] discretizes continuous scattering problems on S^3 via harmonic analysis and discrete polynomials. Avery [23] formulated the generalized Sturmian secular equation for multi-electron systems, using configuration-dependent scaling parameters that are closely related to the β_k eigenvalues of Sec. IX. Neither group previously combined these bases with qubit encoding for quantum simulation—the application explored in Paper 14 [20].

b. Methods. Two Sturmian variants were tested against standard hydrogenic FCI for the He ground state (exact: -2.9037 Ha).

Coulomb Sturmian CI: Each orbital (n, l, m) uses $Z_{\text{eff}} = nk$, where k is a common scaling parameter optimized variationally. The basis is Löwdin-orthogonalized before standard FCI via Slater–Condon rules with full R^k Slater integrals and Gaunt angular coefficients.

Generalized Sturmian CI: Each Slater determinant ν receives its own scaling β_ν from the isoenergetic condition $E_{\text{trial}} = -\beta_\nu^2 Z^2/2 \sum_i 1/n_i^2$. All orbitals within one configuration share $Z_{\text{eff}} = \beta_\nu Z$. A self-consistency loop iterates $E_{\text{trial}} \to \beta_\nu \to (H, S) \to$ generalized eigenvalue $\to E_{\text{new}}$ until convergence.

TABLE I. He ground state energy comparison (exact: -2.9037 Ha). ΔE is energy difference from standard FCI at same $n_{\max}$.

$n_{\max}$	Method	E (Ha)	Error (%)	ΔE (mHa)
2	Standard FCI	−2.8078	3.30	—
2	Coulomb Sturmian	−2.8116	3.17	−3.8
2	Generalized	−2.8249	2.72	−17.1
3	Standard FCI	−2.8172	2.98	—
3	Coulomb Sturmian	−2.8440	2.06	−26.8
3	Generalized	−2.8319	2.47	−14.7

c. Results. The Coulomb Sturmian improves on standard FCI at both basis sizes, with the advantage growing from 0.13 percentage points (pp) at $n_{\max} = 2$ to 0.92 pp at $n_{\max} = 3$. The generalized Sturmian is best at $n_{\max} = 2$ (−17.1 mHa, 2.72% error) but exhibits crossover at $n_{\max} = 3$: it is 12.1 mHa *worse* than the Coulomb variant.

d. β differentiation. The generalized Sturmian at $n_{\max} = 3$ produces six distinct β values ranging from 0.84 (the $1s^2$ configuration) to 2.52 (the $3d^2$ configuration)—a factor 3× spread. Configuration-level differentiation *is* present: compact core configurations receive smaller β (more diffuse orbitals), while high-n configurations receive larger β (more contracted). However, the isoenergetic constraint forces all orbitals within a single configuration to share one $Z_{\text{eff}} = \beta_\nu Z$, removing the n-dependent radial differentiation that the Coulomb Sturmian provides ($Z_{\text{eff}} = nk$). At small basis ($n_{\max} = 2$), between-configuration flexibility dominates and the generalized method wins. At larger basis ($n_{\max} = 3$), the loss of within-configuration orbital flexibility outweighs this gain.

e. Structural finding. The generalized Sturmian's isoenergetic constraint removes n-dependent radial differentiation—a multi-electron analog of the single-electron structural theorem (Theorem 4). In the one-electron case (single-n D-matrix approximation), the shared $SO(4)$ congruence of (H, S) makes the eigenvalues geometry-independent. In the multi-electron case, the constraint $Z_{\text{eff}} = \beta_\nu Z$ (uniform within each configuration) limits the radial flexibility that V_{ee} can exploit, and no variant tested overcomes the fundamental tradeoff between mathematical elegance and radial flexibility.

f. Gaunt sparsity preservation. The Gaunt selection rule sparsity is exactly preserved in both Sturmian variants: the electron repulsion integral (ERI) nonzero counts are identical to the standard hydrogenic basis at each $n_{\max}$ (107 at $n_{\max} = 2$; 3800 at $n_{\max} = 3$; both bases measured, not inherited). This is expected: Gaunt coefficients depend only on (l, m) quantum numbers, not on radial functions or scaling parameters. The result confirms that the angular selection rule structure identified in Paper 7 [3] is *basis-independent*—a property of the angular labels, not of the specific radial representation.

g. Qubit encoding comparison. Track BU-2 compared the Sturmian and standard qubit Hamiltonians (via Löwdin orthogonalization and Jordan–Wigner transformation; see Paper 14 [20] for methodology). At $Q = 10$ qubits ($n_{\max} = 2$), the Sturmian and standard Hamiltonians have identical Pauli term counts (280 each, identity included) but the Sturmian 1-norm is 2.4× higher (26.3 vs. 10.8 Ha). At $Q = 28$ ($n_{\max} = 3$) the counts are again identical (14,047 each), while the 1-norm inflation grows to 3.3× (242 vs. 72.6 Ha). (The standard-encoding figures were re-measured 2026-09-25 on He's physical one-body operator; the earlier 288 / 11.2 Ha and 14,079 / 74 Ha included a non-physical graph hop, Paper 14.) The 1-norm inflation is structural: the Löwdin $S^{-1/2}$ transformation spreads overlap matrix structure into coefficient magnitudes, inflating operator weights throughout the qubit Hamiltonian. Since the Pauli 1-norm controls Trotter error bounds and measurement shot budgets, the Sturmian basis is *disadvantaged* for quantum simulation despite its modest energetic advantages in classical FCI.

h. Negative conclusion. The multi-electron Sturmian direction does not bypass the accuracy ceilings established by the natural geometry hierarchy (Papers 11–17). The investigation yields four structural findings: (a) the single-electron structural theorem extends qualitatively to multi-electron systems—structural constraints on scaling parameters limit radial flexibility regardless of V_{ee} coupling; (b) the convergent dis-

covery with Aquilanti and Avery confirms that S^3 harmonic analysis and Gaunt-based angular algebra are canonical structures, arrived at independently from both directions; (c) Gaunt selection rule sparsity is basis-independent, depending only on the angular quantum numbers, not on radial functions; (d) Löwdin orthogonalization inflates the Pauli 1-norm by 3–5×, making nonorthogonal bases structurally disadvantaged for qubit Hamiltonians.

XIV. $SO(4)$ SELECTION RULES FOR σ-BOND FORMATION

The $SO(4)$ Wigner D-matrix encodes the full angular coupling between atomic orbitals under the bond rotation $e^{i\gamma A_y}$. Two diagonal elements exhibit remarkable constancy.

Theorem 5 (σ-bond selection rules)**.** *For the $n = 2$ representation of $SO(4)$:*

$$D^{(2)}_{(1,0),(1,0)}(\gamma) = 1 \quad \textit{for all } \gamma, \tag{36}$$

$$D^{(2)}_{(0,0),(1,0)}(\gamma) = 0 \quad \textit{for all } \gamma. \tag{37}$$

That is, the bond rotation generated by A_y:

1. *preserves the $2p_0$ orbital ($l = 1, m = 0$) without any attenuation at any bond angle, and*
2. *does not mix the $2s$ orbital ($l = 0, m = 0$) with the $2p_0$ orbital ($l = 1, m = 0$).*

Proof. The D-matrix in the (l, m) basis is computed from the $\mathbf{J}^{(\pm)}$ Clebsch–Gordan decomposition (Eq. (15)) with $j = (n-1)/2 = 1/2$ for $n = 2$. The opposite-sign rotation $d^{1/2}(+\gamma) \otimes d^{1/2}(-\gamma)$ yields, for the $(l' = 1, m' = 0) \to (l = 1, m = 0)$ element:

$$D^{(2)}_{(1,0),(1,0)} = \cos^2(\gamma/2) + \sin^2(\gamma/2) = 1. \tag{38}$$

For the $(l' = 0, m' = 0) \to (l = 1, m = 0)$ element:

$$D^{(2)}_{(0,0),(1,0)} = \cos(\gamma/2)\sin(\gamma/2) - \sin(\gamma/2)\cos(\gamma/2) = 0. \tag{39}$$

Both identities hold for all $\gamma \in [0, \pi]$. □

a. Physical interpretation. The first identity states that a $2p_0$ (σ) orbital is perfectly transmitted through the bond rotation at any internuclear distance—it is a *transparent mode* of the $SO(4)$ bond channel. The second states that the $2s$ orbital cannot scatter into $2p_0$ via the bond rotation alone. Together, these enforce the quantum mechanical selection rule that σ-bonds are formed by p_z orbitals aligned along the bond axis, not by s–p mixing through the Runge–Lenz rotation. The s–p hybridization that creates σ-bonds in the conventional picture arises from the *energy optimization* (variational principle), not from the geometric coupling.

b. Numerical verification. These identities are verified numerically across 10,000 values of $\gamma \in [0.01, \pi - 0.01]$ using the `wigner_D_so4` implementation. The computed values satisfy $|D^{(2)}_{(1,0),(1,0)}(\gamma) - 1| < 10^{-14}$ and $|D^{(2)}_{(0,0),(1,0)}(\gamma)| < 10^{-14}$ at every grid point, and each element matches its closed form from the proof above (`tests/test_paper8_sigma_bond_selection.py`).

XV. CONCLUSION

This paper establishes two classes of results for the bond sphere—the description of a diatomic molecule as a single S^3 with two weighted poles.

a. Positive structural results. The bond sphere geometry is consistent and theoretically motivated. The internuclear distance encodes as a polar angle γ via $\cos\gamma = (p_0^2 - p_R^2)/(p_0^2 + p_R^2)$, and all cross-atom matrix elements are $SO(4)$ Wigner D-matrix elements evaluated at the bond rotation $e^{i\gamma A_y}$ (Eq. (11)). The Runge–Lenz alignment argument (Sec. V) shows that s–p hybridization is a dynamical consequence of the two-center potential. For heteronuclear molecules, the bond sphere becomes an egg—a prolate spheroid whose asymmetry encodes bond polarity (Sec. VI). The kinetic membrane (Sec. VII) is proposed as a combinatorial analog of Bader's bond critical point (unverified; and for LiH its shift runs opposite to Bader's). The $SO(4)$ selection rules (Theorem 5) show that $D^2_{10,10}(\gamma) = 1$ and $D^2_{00,10}(\gamma) = 0$ for all γ: the $2p_0$ orbital is a transparent mode of the bond channel, and s–p mixing requires the variational principle, not the geometric coupling. The single-atom lattice of Papers 0–7 is the $\gamma = 0$ limit; the 18 symbolic proofs and the kinetic scale factor $-1/16$ are unchanged.

b. Negative theorem: Sturmian basis limits. The Sturmian structural theorem (Theorem 4) proves that in the shared-p_0 Sturmian basis, the one-electron (H, S) obey the two-term identity $H_{ij} = -\frac{1}{2}p_0^2 S_{ij} + (1 - \beta_j)V_{ij}$ and carry their entire R-dependence in a single orthogonal $SO(4)$ congruence, making the generalized eigenvalues independent of the bond angle γ and hence of R. In this single-n, shared-p_0 form, binding would require the full R-dependent prolate spheroidal eigenvalue spectrum $\beta_k(R)$; the genuine cross-n Coulomb–Sturmian overlap escapes the theorem and supplies it algebraically (Corollary 2), so what is ruled out is this approximation, not Sturmian molecular binding. For heteronuclear diatomics, no shared p_0 simultaneously represents both atomic length scales (Corollary 3). This theorem was verified computationally across 29 diagnostic versions (v0.9.20–v0.9.34), all of which failed for LiH by the predicted mechanism.

c. Computational path forward. The bond sphere is geometrically consistent but its computational implementation requires either the full LCAO architecture—which preserves atom-specific length scales and produces a vertical binding energy $D_e^{\mathrm{CP}} = 0.110$ Ha at nmax= 3

($\sim$ 20% error vs. experiment, with the R-independent graph-Laplacian kinetic energy giving a monotonically attractive PES with no equilibrium geometry, documented in the companion paper [4])—or a non-Sturmian basis that avoids the $SO(4)$-congruence (geometry-blind) constraint of Theorem 4. The LCAO path is superseded (FCI-M: no equilibrium geometry; the architecture moved to natural geometries); the bond sphere geometry remains a theoretical frame for reading those results.

d. Resolution: the prolate spheroidal lattice. The failure of the single-p_0 Bond Sphere approach points toward the correct solution: rather than forcing both centers onto a shared S^3 with one momentum parameter, each center retains its identity while being embedded in the natural two-center geometry—the prolate spheroid. In prolate spheroidal coordinates (ξ, η, φ), the two-center Coulomb problem separates exactly, and working in the label set and channel structure of these coordinates yields a molecular lattice with R-dependence built into the metric. For H_2^+, this approach achieves $R_{\mathrm{eq}} = 2.001$ bohr (0.21% error) at the finite-difference level, while the spectral Laguerre radial solver reaches machine precision, both with zero fitted parameters [19].

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